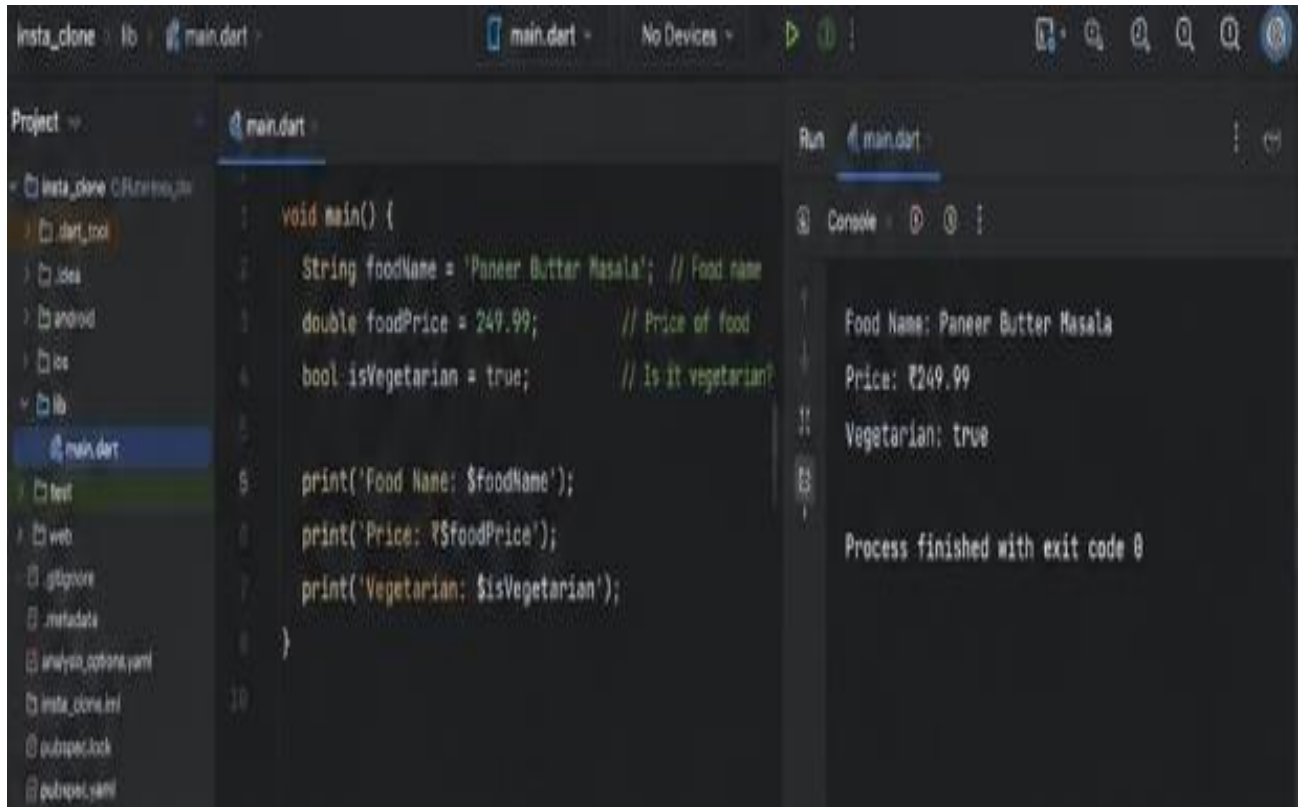


1.

Open main.dart in your Flutter project, declare variables for your favorite food's name (String), its price (double), and whether it's vegetarian (bool), then print all three values to the console using print().

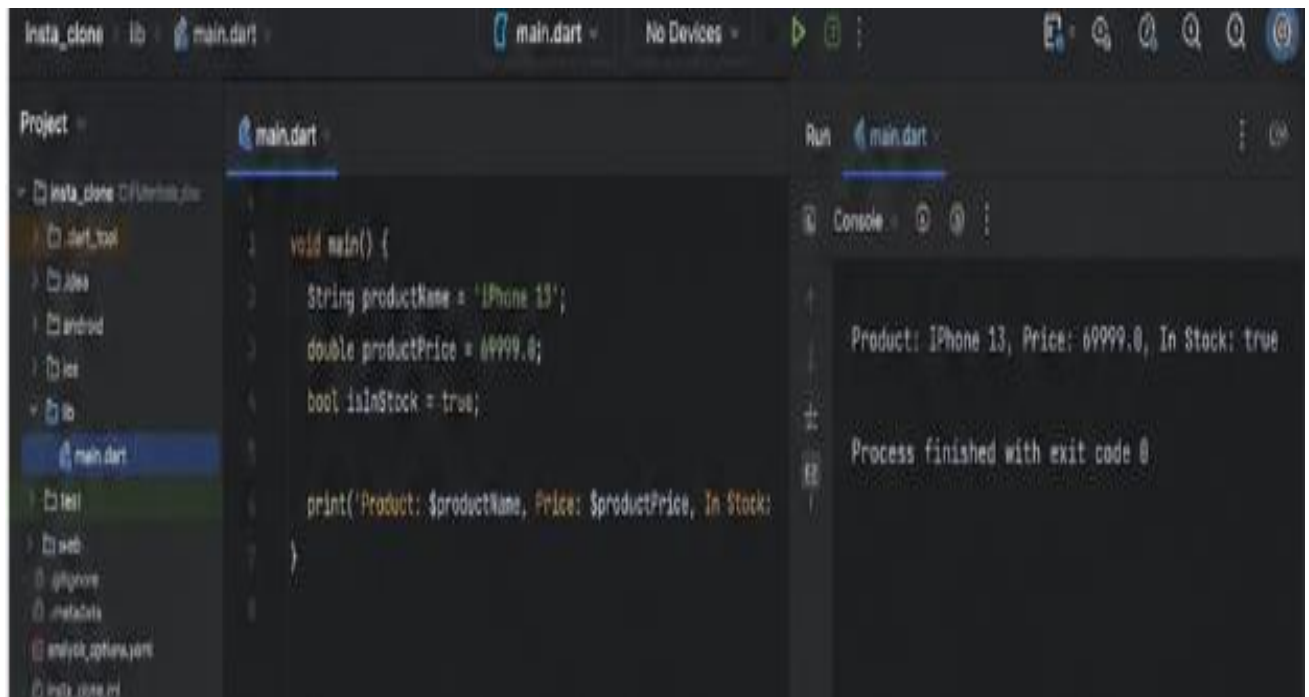


The screenshot shows an IDE window with a project named 'insta_clone'. The file 'main.dart' is open in the editor. The code defines three variables: a String for food name, a double for price, and a bool for vegetarian status. These values are then printed to the console. The console output shows the values: 'Food Name: Paneer Butter Masala', 'Price: ₹249.99', and 'Vegetarian: true'. The process finished with exit code 0.

```
void main() {  
  String foodName = 'Paneer Butter Masala'; // Food name  
  double foodPrice = 249.99; // Price of food  
  bool isVegetarian = true; // Is it vegetarian?  
  
  print('Food Name: $foodName');  
  print('Price: ₹$foodPrice');  
  print('Vegetarian: $isVegetarian');  
}
```

Food Name: Paneer Butter Masala
Price: ₹249.99
Vegetarian: true
Process finished with exit code 0

2. Write Dart code in main.dart to declare variables for a Flipkart product: productName (String), productPrice (double), and isInStock (bool). Print a message like 'Product: iPhone 13, Price: 69999.0, In Stock: true' using string interpolation.



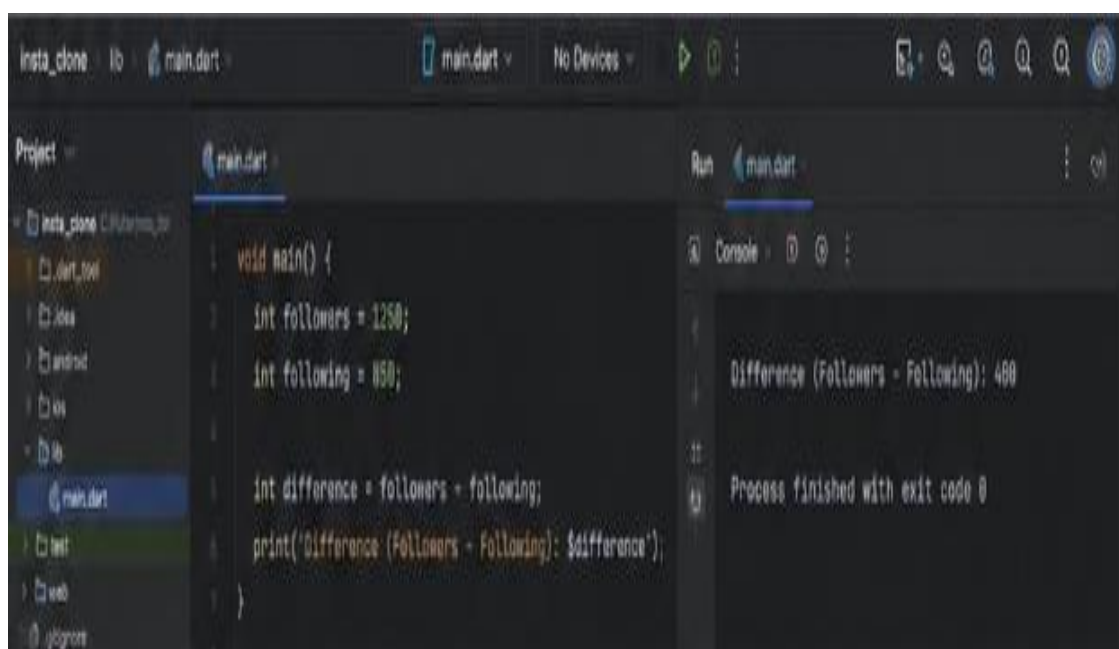
The screenshot shows an IDE with a project named 'insta_clone'. The file 'main.dart' is open, containing the following Dart code:

```
void main() {  
  String productName = 'iPhone 13';  
  double productPrice = 69999.0;  
  bool isInStock = true;  
  
  print('Product: $productName, Price: $productPrice, In Stock: $isInStock');  
}
```

The console output on the right shows the result of running the code:

```
Product: iPhone 13, Price: 69999.0, In Stock: true  
  
Process finished with exit code 0
```

3. Declare two integer variables in main.dart: followers and following for a user profile (like Instagram). Calculate and print the difference (followers - following) to the console.



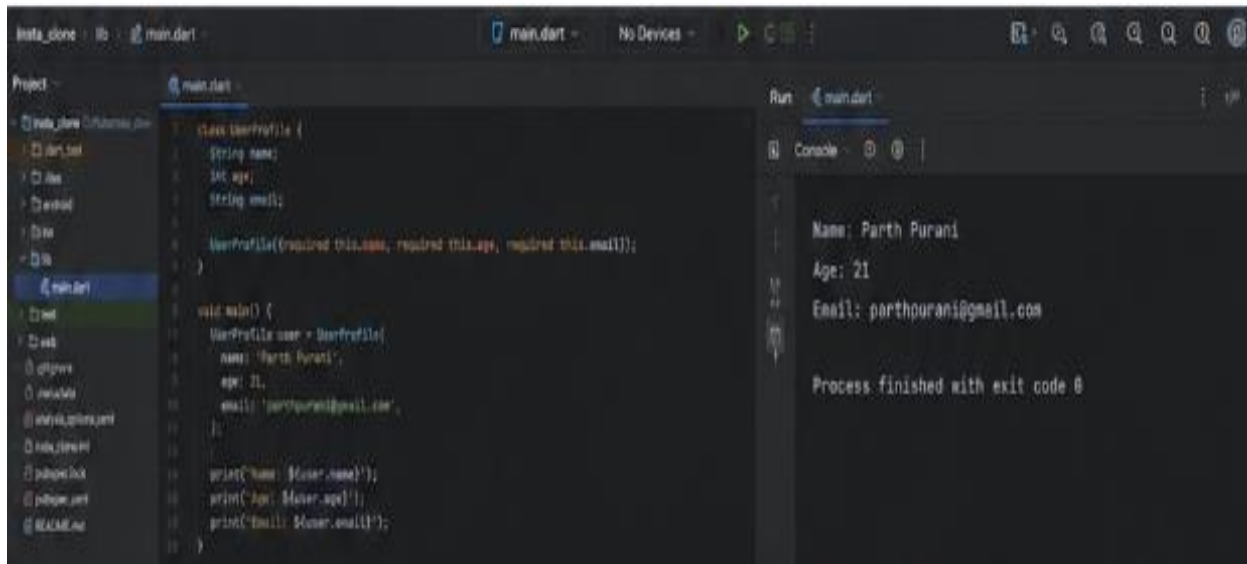
The screenshot shows the same IDE with 'main.dart' open, containing the following Dart code:

```
void main() {  
  int followers = 1250;  
  int following = 850;  
  
  int difference = followers - following;  
  print('Difference (Followers - Following): $difference');  
}
```

The console output on the right shows the result of running the code:

```
Difference (Followers - Following): 400  
  
Process finished with exit code 0
```

4. Refactor your main.dart code to group all user details (name, age, email) into a single UserProfile class with appropriate data types. Create an object of UserProfile, assign values, and print each property.



The screenshot shows an IDE with a project named 'insta_clone'. The file 'main.dart' is open and contains the following code:

```
class UserProfile {
  String name;
  int age;
  String email;

  UserProfile(required this.name, required this.age, required this.email);
}

void main() {
  UserProfile user = UserProfile(
    name: 'Parth Purani',
    age: 21,
    email: 'parthpurani@gmail.com',
  );

  print('Name: $user.name');
  print('Age: $user.age');
  print('Email: $user.email');
}
```

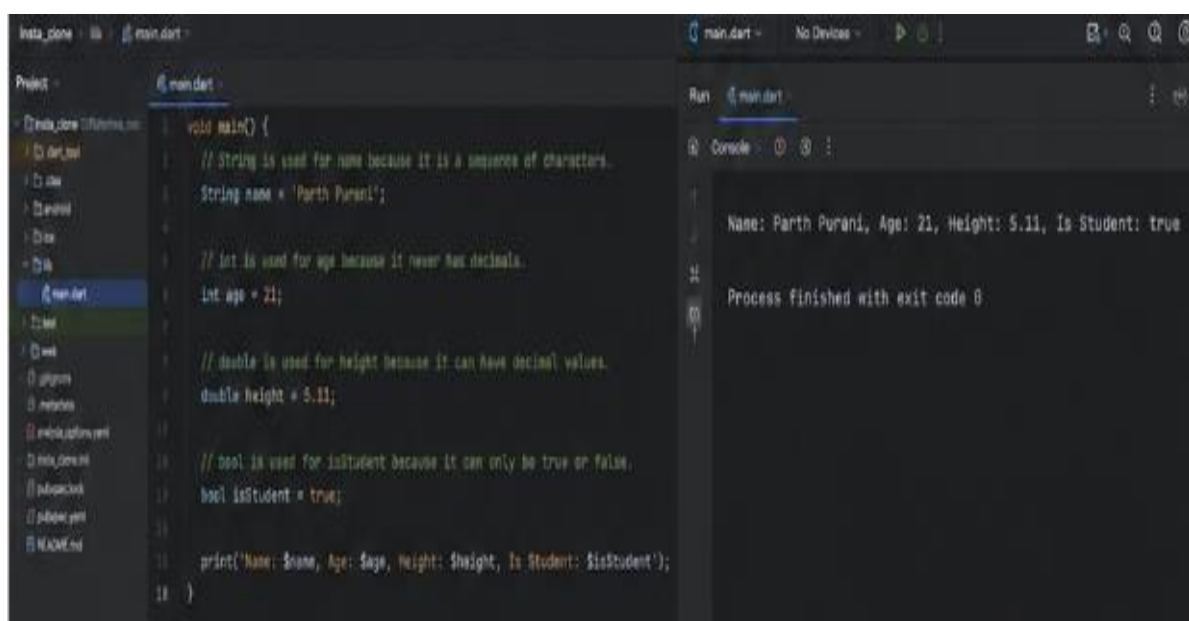
The console output on the right shows:

```
Name: Parth Purani
Age: 21
Email: parthpurani@gmail.com

Process finished with exit code 0
```

5. Rewrite your variable declarations in main.dart to use explicit types instead of var, and add a comment above each variable explaining why that data type is appropriate.

Hint: For example, use int for age because it never has decimals.



The screenshot shows the same IDE with the updated 'main.dart' file. The code is as follows:

```
void main() {
  // String is used for name because it is a sequence of characters.
  String name = 'Parth Purani';

  // int is used for age because it never has decimals.
  int age = 21;

  // double is used for height because it can have decimal values.
  double height = 5.11;

  // bool is used for isStudent because it can only be true or false.
  bool isStudent = true;

  print('Name: $name, Age: $age, Height: $height, Is Student: $isStudent');
}
```

The console output on the right shows:

```
Name: Parth Purani, Age: 21, Height: 5.11, Is Student: true

Process finished with exit code 0
```

